Experiment 22

MySQL Queries for Train Ticket Reservation System with Social Network Notifications

Aim

To manage passenger details, trains, ticket reservations, fare calculations, booking status, and stored social network notification messages using MySQL.

Software Required

MySQL Workbench or MySQL Command Line Client.

SQL Program

Step 1: Create database

```
CREATE DATABASE TrainReservationDB;
USE TrainReservationDB;
```

Step 2: Create tables

```
CREATE TABLE Passenger (
    passenger_id INT PRIMARY KEY,
    passenger_name VARCHAR(50),
    phone VARCHAR(15),
    social_id VARCHAR(50)
);

CREATE TABLE Train (
    train_id INT PRIMARY KEY,
    train_name VARCHAR(50),
    source VARCHAR(50),
    destination VARCHAR(50),
    fare DECIMAL(8,2)
);

CREATE TABLE Ticket (
    ticket_id INT PRIMARY KEY,
    passenger_id INT,
    train_id INT,
    journey_date DATE,
    seats INT,
    status VARCHAR(20),
    FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
    FOREIGN KEY (train_id) REFERENCES Train(train_id)
);

CREATE TABLE SocialNotification (
    notification_id INT PRIMARY KEY,
    passenger_id INT,
    message VARCHAR(200),
    sent_date DATE,
    FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id)
);
```

Step 3: Insert sample data

```
INSERT INTO Passenger VALUES
(1, 'Ravi', '9876543210', 'ravi_social'),
(2, 'Priya', '9876501234', 'priya_social'),
(3, 'Arun', '9876512345', 'arun_social');

INSERT INTO Train VALUES
(101, 'Chennai Express', 'Chennai', 'Bangalore', 500),
(102, 'Delhi Express', 'Delhi', 'Mumbai', 800),
(103, 'Coimbatore Express', 'Chennai', 'Coimbatore', 450);

INSERT INTO Ticket VALUES
(1001, 1, 101, '2026-11-05', 2, 'Confirmed'),
(1002, 2, 102, '2026-11-08', 1, 'Confirmed'),
(1003, 3, 103, '2026-11-10', 1, 'Pending');

INSERT INTO SocialNotification VALUES
(1, 1, 'Your Chennai Express ticket is confirmed.', '2026-10-24'),
(2, 2, 'Your Delhi Express ticket is confirmed.', '2026-10-24');
```

Queries

Display trains

```
SELECT * FROM Train;
```

Search a route

```
SELECT * FROM Train WHERE source = 'Chennai' AND destination = 'Bangalore';
```

Display passenger ticket details

```
SELECT p.passenger_name, t.train_name, t.source, t.destination,
       tk.journey_date, tk.seats, tk.status
FROM Passenger p
JOIN Ticket tk ON p.passenger_id = tk.passenger_id
JOIN Train t ON tk.train_id = t.train_id;
```

Calculate total fare

```
SELECT p.passenger_name, t.train_name, tk.seats, t.fare,
       tk.seats * t.fare AS total_fare
FROM Passenger p
JOIN Ticket tk ON p.passenger_id = tk.passenger_id
JOIN Train t ON tk.train_id = t.train_id;
```

Update ticket status

```
UPDATE Ticket SET status = 'Confirmed' WHERE ticket_id = 1003;
```

Insert a notification

```
INSERT INTO SocialNotification VALUES
(3, 3, 'Your Coimbatore Express ticket is confirmed.', '2026-10-24');
```

Display notifications

```
SELECT p.passenger_name, p.social_id, s.message, s.sent_date
FROM Passenger p
JOIN SocialNotification s ON p.passenger_id = s.passenger_id;
```

Count confirmed tickets

```
SELECT COUNT(*) AS confirmed_tickets FROM Ticket WHERE status = 'Confirmed';
```

Expected Output:

```
mysql> CREATE DATABASE TrainReservationDB;
Query OK, 1 row affected (0.05 sec)

mysql> USE TrainReservationDB;
Database changed
mysql> CREATE TABLE Passenger (
    ->     passenger_id INT PRIMARY KEY,
    ->     passenger_name VARCHAR(50),
    ->     phone VARCHAR(15),
    ->     social_id VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql>
mysql> CREATE TABLE Train (
    ->     train_id INT PRIMARY KEY,
    ->     train_name VARCHAR(50),
    ->     source VARCHAR(50),
    ->     destination VARCHAR(50),
    ->     fare DECIMAL(8,2)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Ticket (
    ->     ticket_id INT PRIMARY KEY,
    ->     passenger_id INT,
    ->     train_id INT,
    ->     journey_date DATE,
    ->     seats INT,
    ->     status VARCHAR(20),
```




```
mysql>
mysql> CREATE TABLE SocialNotification (
    ->     notification_id INT PRIMARY KEY,
    ->     passenger_id INT,
    ->     message VARCHAR(200),
    ->     sent_date DATE,
    ->     FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id)
    -> );
Query OK, 0 rows affected (0.12 sec)

mysql> INSERT INTO Passenger VALUES
    -> (1, 'Ravi', '9876543210', 'ravi_social'),
    -> (2, 'Priya', '9876501234', 'priya_social'),
    -> (3, 'Arun', '9876512345', 'arun_social');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Train VALUES
    -> (101, 'Chennai Express', 'Chennai', 'Bangalore', 500),
    -> (102, 'Delhi Express', 'Delhi', 'Mumbai', 800),
    -> (103, 'Coimbatore Express', 'Chennai', 'Coimbatore', 450);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Ticket VALUES
    -> (1001, 1, 101, '2026-11-05', 2, 'Confirmed'),
    -> (1002, 2, 102, '2026-11-08', 1, 'Confirmed'),
    -> (1003, 3, 103, '2026-11-10', 1, 'Pending');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

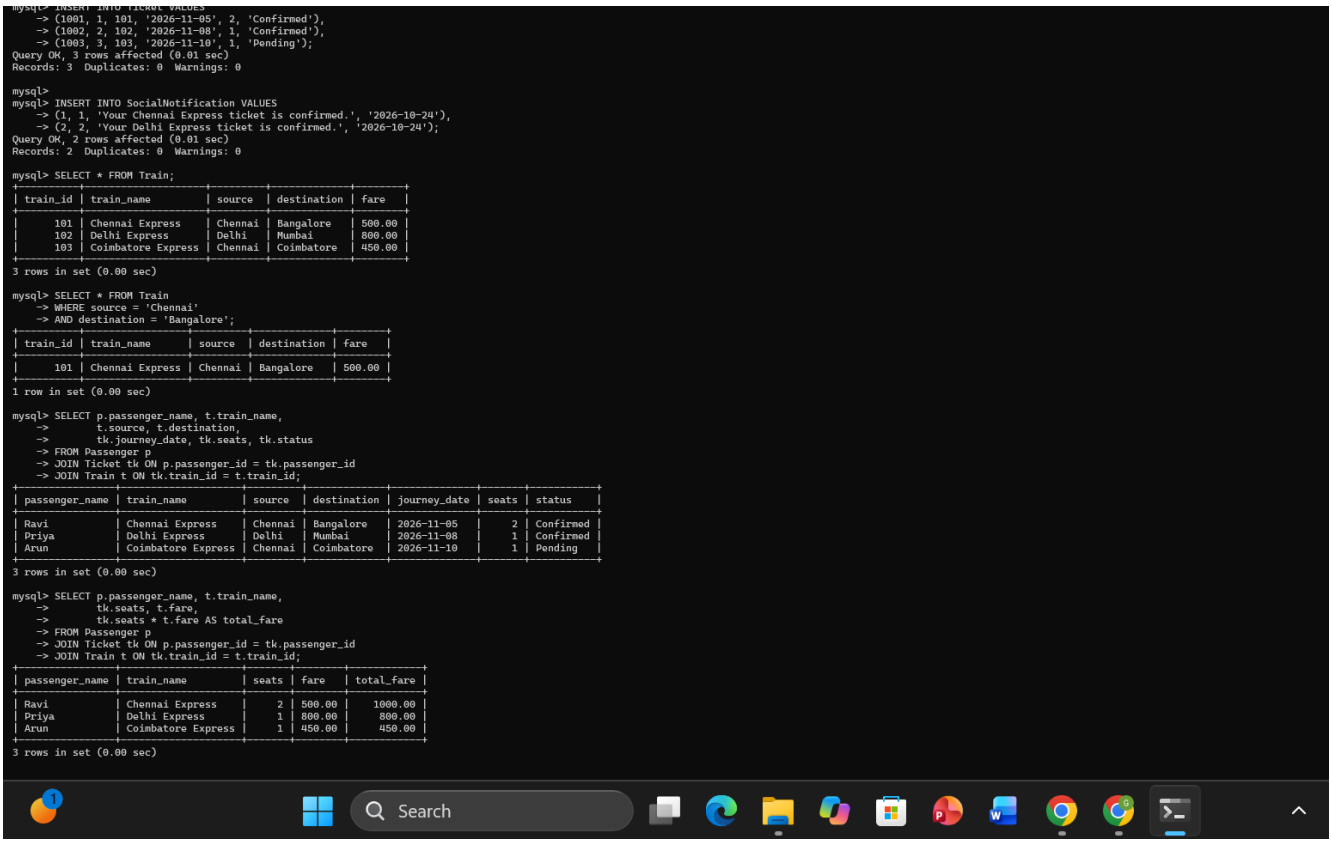


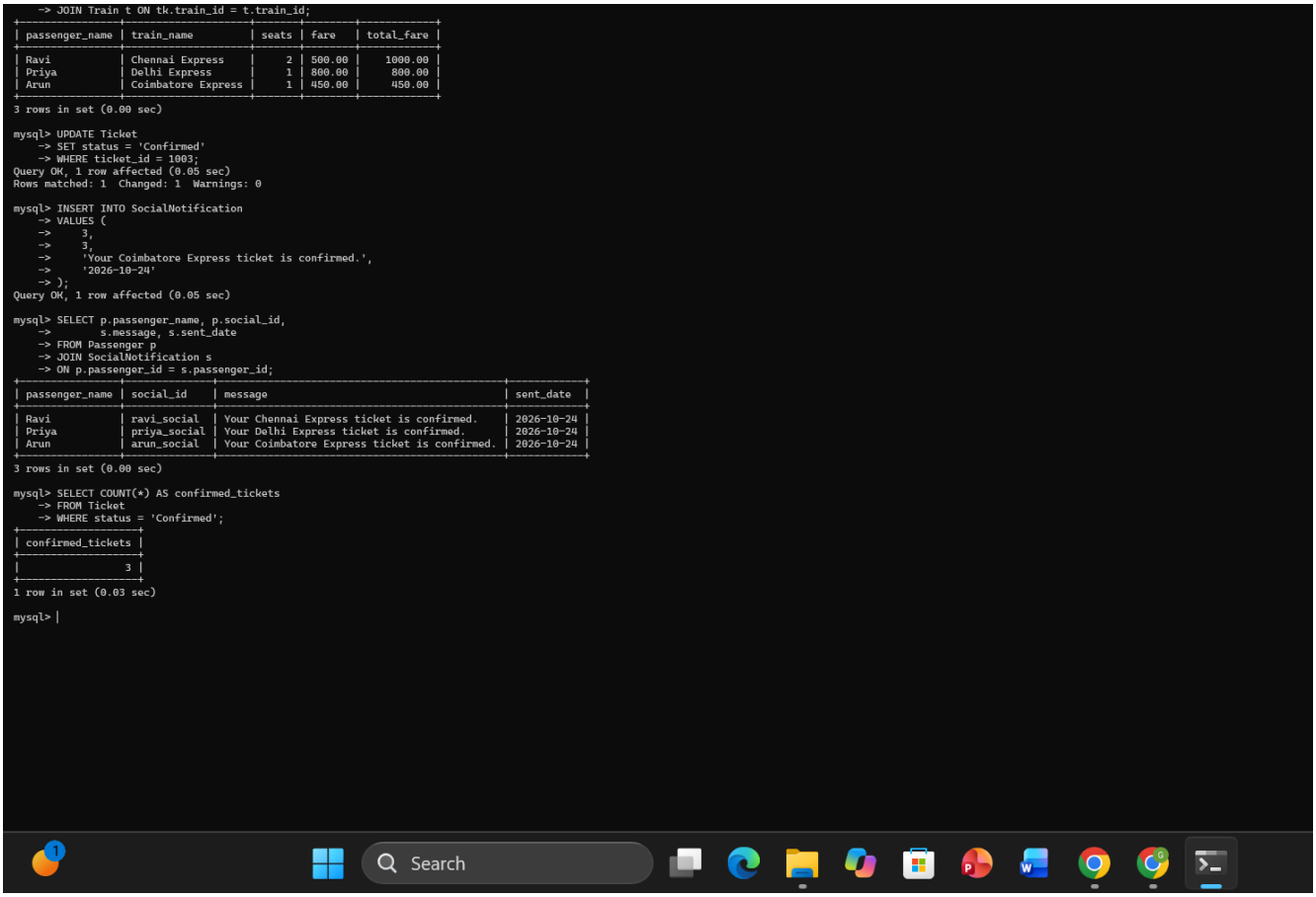
```
mysql>
mysql> INSERT INTO Ticket VALUES
    -> (1001, 1, 101, '2026-11-05', 2, 'Confirmed'),
    -> (1002, 2, 102, '2026-11-08', 1, 'Confirmed'),
    -> (1003, 3, 103, '2026-11-10', 1, 'Pending');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO SocialNotification VALUES
    -> (1, 1, 'Your Chennai Express ticket is confirmed.', '2026-10-24'),
    -> (2, 2, 'Your Delhi Express ticket is confirmed.', '2026-10-24');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Train;
+----------+--------------------+---------+-------------+--------+
| train_id | train_name         | source  | destination | fare   |
+----------+--------------------+---------+-------------+--------+
|      101 | Chennai Express    | Chennai | Bangalore   | 500.00 |
|      102 | Delhi Express      | Delhi   | Mumbai      | 800.00 |
|      103 | Coimbatore Express | Chennai | Coimbatore  | 450.00 |
+----------+--------------------+---------+-------------+--------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Train
    -> WHERE source = 'Chennai'
    -> AND destination = 'Bangalore';
+----------+------------------+---------+-------------+--------+
| train_id | train_name       | source  | destination | fare   |
+----------+------------------+---------+-------------+--------+
|      101 | Chennai Express | Chennai | Bangalore   | 500.00 |
```





Observation

The tables and queries manage train data, passenger records, bookings, fares, status updates, and notification messages.

Result

Thus, MySQL queries for a train reservation system with stored social network notifications were implemented successfully.

Note

Messages are stored in the database only. Sending them through a social network requires that platform's API and authentication.